CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 12: Mid-point and Its Converse

Every formula and result of the chapter on one sheet

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|---|---|---|
| 1 | Mid-point theorem | $DE \parallel BC$ and $DE = \frac{1}{2}BC$ |
| 2 | Converse | A line through the mid-point of one side, parallel to another, bisects the third side |
| 3 | Mid-point triangle, perimeter | $\frac{1}{2}\times$ perimeter of the triangle |
| 4 | Mid-point triangle, area | $\frac{1}{4}\times$ area of the triangle |
| 5 | Mid-points of a quadrilateral | Form a parallelogram |
| 6 | Sides of that parallelogram | $\frac{1}{2}d_1$ and $\frac{1}{2}d_2$ |
| 7 | Perimeter of that parallelogram | $d_1 + d_2$ |
| 8 | Trapezium, mid-points of the non-parallel sides | $EF = \frac{1}{2}(AB + DC)$ |
| 9 | Trapezium, mid-points of the diagonals | $\frac{1}{2}(AB - DC)$ |
| 10 | Equal intercept theorem | Equal intercepts on one transversal give equal intercepts on every transversal |
| 11 | Rectangle | Mid-point figure is a rhombus |
| 12 | Rhombus | Mid-point figure is a rectangle |